

Лабораторная работа № 6

Настройка пропускной способности глобальной сети с помощью Token Bucket Filter

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Информация

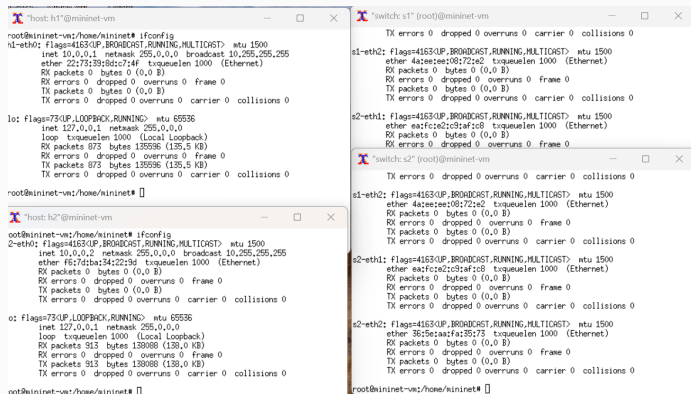
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Основной целью работы является знакомство с принципами работы дисциплины очереди Token Bucket Filter, которая формирует входящий/исходящий трафик для ограничения пропускной способности, а также получение навыков моделирования и исследования поведения трафика посредством проведения интерактивного и воспроизводимого экспериментов в Mininet.

1. Задайте топологию, состоящую из двух хостов и двух коммутаторов с назначенной по умолчанию mininet сетью 10.0.0.0/8.
2. Проведите интерактивные эксперименты по ограничению пропускной способности сети с помощью TBF в эмулируемой глобальной сети.
3. Самостоятельно реализуйте воспроизводимые эксперимент по применению TBF для ограничения пропускной способности. Постройте соответствующие графики.

Выполнение лабораторной работы



```
"host: h1"@mininet-vm
root@mininet-vm:/home/mininet# ifconfig
h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.1 netmask 255.0.0.0 broadcast 10.255.255.255
    ether 22:73:3b:d5:74:f4 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 873 bytes 135596 (135.5 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 873 bytes 135596 (135.5 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet#

"host: h2"@mininet-vm
root@mininet-vm:/home/mininet# ifconfig
2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.2 netmask 255.0.0.0 broadcast 10.255.255.255
    ether f6:7d:ba:34:22:3d txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 913 bytes 138088 (138.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 913 bytes 138088 (138.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet#

"switch: s1" (root@mininet-vm
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

s1-eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether 4a:ee:ee:08:72:e2 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

s2-eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether ea:fc:e2:c9:af:c8 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0

"switch: s2" (root@mininet-vm
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

s1-eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether 4a:ee:ee:08:72:e2 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

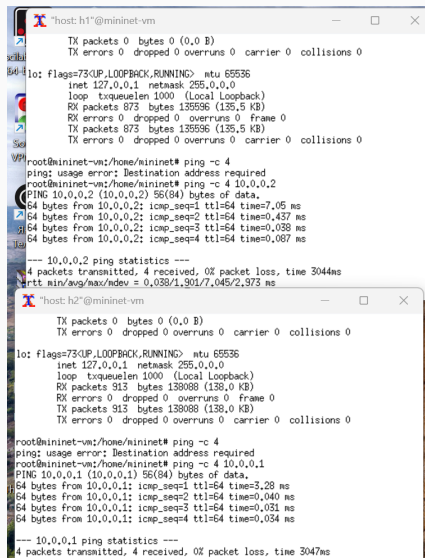
s2-eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether ea:fc:e2:c9:af:c8 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

s2-eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether 36:5e:aa:fa:35:73 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet#
```

Figure 1: Задание топологии

Выполнение лабораторной работы



```
"host: h1"@mininet-vm
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
loop txqueuelen 1000 (Local Loopback)
RX packets 873 bytes 135596 (135.5 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 873 bytes 135596 (135.5 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet# ping -c 4
ping: usage error: Destination address required
root@mininet-vm:/home/mininet# ping -c 4 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=7.05 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.437 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.038 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.087 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3044ms
rtt min/avg/max/ndev = 0.038/1.901/7.045/2.973 ms

"host: h2"@mininet-vm
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
loop txqueuelen 1000 (Local Loopback)
RX packets 913 bytes 138088 (138.0 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 913 bytes 138088 (138.0 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet# ping -c 4
ping: usage error: Destination address required
root@mininet-vm:/home/mininet# ping -c 4 10.0.0.1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=3.28 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0.040 ns
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=0.031 ns
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0.034 ns

--- 10.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3047ms
```

Figure 2: ifconfig на хостах

Выполнение лабораторной работы

```
"host: h1"@mininet-vm
rtt min/avg/max/ndev = 0,038/1,301/7,045/2,973 ms
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
iperf3: error - unable to connect to server: Connection refused
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 39612 connected to 10.0.0.2 port 5201
[ ID] Interval           Transfer     Bitrate      Retr    Cwnd
[ 7] 0.00-1.00 sec      5.59 GBytes  48.0 Gbits/sec    0     4.39 MBytes
[ 7] 1.00-2.00 sec      5.71 GBytes  49.1 Gbits/sec    0     4.39 MBytes
[ 7] 2.00-3.00 sec      5.53 GBytes  47.5 Gbits/sec   47     3.49 MBytes
[ 7] 3.00-4.00 sec      5.09 GBytes  43.7 Gbits/sec    1     3.49 MBytes
[ 7] 4.00-5.00 sec      4.89 GBytes  42.0 Gbits/sec    0     3.50 MBytes
[ 7] 5.00-6.00 sec      5.08 GBytes  43.6 Gbits/sec    1     3.50 MBytes
[ 7] 6.00-7.00 sec      5.62 GBytes  48.2 Gbits/sec    0     3.50 MBytes
[ 7] 7.00-8.00 sec      5.70 GBytes  48.9 Gbits/sec   47     2.90 MBytes
[ 7] 8.00-9.00 sec      5.50 GBytes  47.3 Gbits/sec    0     2.90 MBytes
[ 7] 9.00-10.00 sec     5.55 GBytes  47.6 Gbits/sec    0     2.90 MBytes
-----
[ ID] Interval           Transfer     Bitrate      Retr    sender
[ 7] 0.00-10.00 sec     54.3 GBytes  46.6 Gbits/sec    96      receiver
[ 7] 0.00-10.00 sec     54.3 GBytes  46.6 Gbits/sec

"host: h2"@mininet-vm
Server listening on 5201
-----
Accepted connection from 10.0.0.1, port 39608
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 39612
[ ID] Interval           Transfer     Bitrate
[ 7] 0.00-1.00 sec      5.59 GBytes  48.0 Gbits/sec
[ 7] 1.00-2.00 sec      5.71 GBytes  49.0 Gbits/sec
[ 7] 2.00-3.00 sec      5.53 GBytes  47.5 Gbits/sec
[ 7] 3.00-4.00 sec      5.09 GBytes  43.7 Gbits/sec
[ 7] 4.00-5.00 sec      4.89 GBytes  42.0 Gbits/sec
[ 7] 5.00-6.00 sec      5.08 GBytes  43.6 Gbits/sec
[ 7] 6.00-7.00 sec      5.62 GBytes  48.2 Gbits/sec
[ 7] 7.00-8.00 sec      5.70 GBytes  48.9 Gbits/sec
[ 7] 8.00-9.00 sec      5.50 GBytes  47.3 Gbits/sec
[ 7] 9.00-10.00 sec     5.55 GBytes  47.6 Gbits/sec
[ 7] 10.00-10.00 sec     3.62 MBytes  52.4 Gbits/sec
-----
[ ID] Interval           Transfer     Bitrate
[ 7] 0.00-10.00 sec     54.3 GBytes  46.6 Gbits/sec
-----
Server listening on 5201
"
```

Figure 3: Запуск iperf3 на хостах

Выполнение лабораторной работы

```
"host: h1"@mininet-vm
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.087 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3044ms
rtt min/avg/max/ndev = 0.038/1.901/7.045/2.973 ms
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
iperf3: error - unable to connect to server: Connection refused
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 39612 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate      Retr  Cwnd
[ 7] 0.00-1.00    sec  5.59 GBytes  48.0 Gbits/sec  0  4.39 MBytes
[ 7] 1.00-2.00    sec  5.71 GBytes  49.1 Gbits/sec  0  4.39 MBytes
[ 7] 2.00-3.00    sec  5.53 GBytes  47.5 Gbits/sec  47  3.49 MBytes
[ 7] 3.00-4.00    sec  5.09 GBytes  43.7 Gbits/sec  1  3.49 MBytes
[ 7] 4.00-5.00    sec  4.89 GBytes  42.0 Gbits/sec  0  3.50 MBytes
[ 7] 5.00-6.00    sec  5.08 GBytes  43.6 Gbits/sec  1  3.50 MBytes
[ 7] 6.00-7.00    sec  5.62 GBytes  48.2 Gbits/sec  0  3.50 MBytes
[ 7] 7.00-8.00    sec  5.70 GBytes  48.9 Gbits/sec  47  2.90 MBytes
[ 7] 8.00-9.00    sec  5.50 GBytes  47.3 Gbits/sec  0  2.90 MBytes
[ 7] 9.00-10.00   sec  5.55 GBytes  47.6 Gbits/sec  0  2.90 MBytes
-----
[ ID] Interval      Transfer    Bitrate      Retr  sender receiver
[ 7] 0.00-10.00   sec  54.3 GBytes  46.6 Gbits/sec  96
[ 7] 0.00-10.00   sec  54.3 GBytes  46.6 Gbits/sec

iperf Done.
root@mininet-vm:/home/mininet# sudo tc qdisc add dev hi-eth0 root tbf rate 10gb
it burst 5000000 limit 15000000
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 53860 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate      Retr  Cwnd
[ 7] 0.00-1.00    sec  1005 MBytes  8.43 Gbits/sec  0  3.37 MBytes
[ 7] 1.00-2.00    sec  1.10 GBytes  9.45 Gbits/sec  0  4.37 MBytes
[ 7] 2.00-3.00    sec  1.11 GBytes  9.58 Gbits/sec  135  4.59 MBytes
[ 7] 3.00-4.00    sec  996 MBytes  8.34 Gbits/sec  1260  3.30 MBytes
[ 7] 4.00-5.00    sec  639 MBytes  5.37 Gbits/sec  0  3.33 MBytes
[ 7] 5.00-6.00    sec  725 MBytes  6.08 Gbits/sec  45  2.44 MBytes
[ 7] 6.00-7.00    sec  1.11 GBytes  9.56 Gbits/sec  0  2.53 MBytes
[ 7] 7.00-8.00    sec  1.11 GBytes  9.57 Gbits/sec  0  2.75 MBytes
[ 7] 8.00-9.00    sec  1.11 GBytes  9.52 Gbits/sec  0  3.07 MBytes
[ 7] 9.00-10.00   sec  931 MBytes  7.81 Gbits/sec  0  3.08 MBytes
-----
[ ID] Interval      Transfer    Bitrate      Retr  sender receiver
[ 7] 0.00-10.00   sec  9.75 GBytes  8.37 Gbits/sec  1440
[ 7] 0.00-10.00   sec  9.74 GBytes  8.36 Gbits/sec

"host: h2"@mininet-vm
Accepted connection from 10.0.0.1, port 39608
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 39612
[ ID] Interval      Transfer    Bitrate
[ 7] 0.00-1.00    sec  5.59 GBytes  48.0 Gbits/sec
[ 7] 1.00-2.00    sec  5.71 GBytes  49.0 Gbits/sec
[ 7] 2.00-3.00    sec  5.53 GBytes  47.5 Gbits/sec
[ 7] 3.00-4.00    sec  5.09 GBytes  43.7 Gbits/sec
[ 7] 4.00-5.00    sec  4.89 GBytes  42.0 Gbits/sec
[ 7] 5.00-6.00    sec  5.08 GBytes  43.6 Gbits/sec
[ 7] 6.00-7.00    sec  5.62 GBytes  48.2 Gbits/sec
[ 7] 7.00-8.00    sec  5.70 GBytes  48.9 Gbits/sec
[ 7] 8.00-9.00    sec  5.50 GBytes  47.3 Gbits/sec
[ 7] 9.00-10.00   sec  5.55 GBytes  47.6 Gbits/sec
[ 7] 10.00-10.00  sec  3.62 MBytes  52.4 Gbits/sec
-----
[ ID] Interval      Transfer    Bitrate
[ 7] 0.00-10.00   sec  54.3 GBytes  46.6 Gbits/sec
receiver

Server listening on 5201

"Ciperf3: interrupt - the server has terminated
root@mininet-vm:/home/mininet# iperf3 -s
warning: this system does not seem to support IPv6 - trying IPv4

Server listening on 5201

Accepted connection from 10.0.0.1, port 53850
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 53860
[ ID] Interval      Transfer    Bitrate
[ 7] 0.00-1.00    sec  994 MBytes  8.34 Gbits/sec
[ 7] 1.00-2.00    sec  1.10 GBytes  9.45 Gbits/sec
[ 7] 2.00-3.00    sec  1.11 GBytes  9.58 Gbits/sec
[ 7] 3.00-4.00    sec  996 MBytes  8.33 Gbits/sec
[ 7] 4.00-5.00    sec  638 MBytes  5.36 Gbits/sec
[ 7] 5.00-6.00    sec  727 MBytes  6.10 Gbits/sec
[ 7] 6.00-7.00    sec  1.11 GBytes  9.56 Gbits/sec
[ 7] 7.00-8.00    sec  1.11 GBytes  9.56 Gbits/sec
[ 7] 8.00-9.00    sec  1.11 GBytes  9.53 Gbits/sec
[ 7] 9.00-10.00   sec  930 MBytes  7.80 Gbits/sec
[ 7] 10.00-10.00  sec  1.37 MBytes  4.13 Gbits/sec
-----
[ ID] Interval      Transfer    Bitrate
[ 7] 0.00-10.00   sec  9.74 GBytes  8.36 Gbits/sec
receiver

Server listening on 5201
n
```

Figure 4: Ограничение скорости на конечных хостах

Выполнение лабораторной работы

```
"switch: s1" (root)@mininet-vm
s1-eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
ether 4a:ee:ee:08:72:e2 txqueuelen 1000 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

s2-eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
ether ea:fc:e2:c9:af:c8 txqueuelen 1000 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

s2-eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
ether 36:5e:aa:fa:35:73 txqueuelen 1000 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 root tbf rate
it burst 5000000 limit 15000000

"host: h1" @mininet-vm
Connecting to host 10.0.0.1 port 5201
[ 7] local 10.0.0.1 port 5201 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer   Bitrate   Retr   Cwnd
[ 7] 0.00-1.00 sec    396 MBytes 8.36 Gbits/sec    64    2.45 MBytes
[ 7] 1.00-2.01 sec    572 MBytes 4.75 Gbits/sec     0    2.49 MBytes
[ 7] 2.01-3.02 sec    429 MBytes 3.55 Gbits/sec   107    2.57 MBytes
[ 7] 3.02-4.00 sec    439 MBytes 3.77 Gbits/sec   225    1.96 MBytes
[ 7] 4.00-5.00 sec    531 MBytes 4.45 Gbits/sec     0    2.12 MBytes
[ 7] 5.00-6.00 sec    395 MBytes 3.32 Gbits/sec   158    2.16 MBytes
[ 7] 6.00-7.00 sec    434 MBytes 3.64 Gbits/sec    45    1.58 MBytes
[ 7] 7.00-8.00 sec    275 MBytes 2.30 Gbits/sec    90    1.68 MBytes
[ 7] 8.00-9.01 sec    362 MBytes 3.00 Gbits/sec     0    1.77 MBytes
[ 7] 9.01-10.00 sec   754 MBytes 6.42 Gbits/sec   135    1.80 MBytes
-----
[ ID] Interval      Transfer   Bitrate   Retr   sender
[ 7] 0.00-10.00 sec    5.07 GBytes 4.35 Gbits/sec    824    receiver
[ 7] 0.00-10.00 sec    5.06 GBytes 4.34 Gbits/sec
iperf Done.
root@mininet-vm:/home/mininet#
```

host: h2" @mininet-vm

```
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 42296
[ ID] Interval      Transfer   Bitrate
[ 7] 0.00-1.00 sec    386 MBytes 8.27 Gbits/sec
[ 7] 1.00-2.01 sec    572 MBytes 4.75 Gbits/sec
[ 7] 2.01-3.03 sec    429 MBytes 3.55 Gbits/sec
[ 7] 3.03-4.00 sec    439 MBytes 3.77 Gbits/sec
[ 7] 4.00-5.00 sec    532 MBytes 4.46 Gbits/sec
[ 7] 5.00-6.00 sec    396 MBytes 3.32 Gbits/sec
[ 7] 6.00-7.00 sec    433 MBytes 3.63 Gbits/sec
[ 7] 7.00-8.00 sec    274 MBytes 2.29 Gbits/sec
[ 7] 8.00-9.01 sec    363 MBytes 3.01 Gbits/sec
[ 7] 9.01-10.00 sec    754 MBytes 6.42 Gbits/sec
[ 7] 10.00-10.00 sec    764 KBytes 5.25 Gbits/sec
-----
[ ID] Interval      Transfer   Bitrate
[ 7] 0.00-10.00 sec    5.06 GBytes 4.34 Gbits/sec
Server listening on 5201
```



Figure 5: Ограничение скорости на коммутаторах

Выполнение лабораторной работы

```
"switch: s1" (root)@mininet-vm
s2-eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether ea:fc:e2:c3:af:c8 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

s2-eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether 36:5e:aa:fa:35:73 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 root tbf rate
it burst 5000000 limit 15000000
root@mininet-vm:/home/mininet# sudo tc qdisc del dev s1-eth2 root
root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 root handle 1:
m delay 10ms
root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 parent 1: hand
: tbf rate 2gbit burst 1000000 limit 2000000
-----
"host: h1" @mininet-vm
connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 57956 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer   Bitrate   Retr  Cwnd
[ 7] 0.00-1.00      sec 60.6 MBytes  508 Mbits/sec  74  1.34 MBytes
[ 7] 1.00-2.04      sec 48.8 MBytes  392 Mbits/sec  37  1.40 MBytes
[ 7] 2.04-3.00      sec 26.2 MBytes  230 Mbits/sec  0  1.41 MBytes
[ 7] 3.00-4.00      sec 53.8 MBytes  452 Mbits/sec  0  1.44 MBytes
[ 7] 4.00-5.00      sec 37.5 MBytes  314 Mbits/sec  0  1.46 MBytes
[ 7] 5.00-6.00      sec 43.8 MBytes  367 Mbits/sec  7  1.48 MBytes
[ 7] 6.00-7.07      sec 27.5 MBytes  215 Mbits/sec  0  1.49 MBytes
[ 7] 7.07-8.06      sec 40.0 MBytes  338 Mbits/sec  0  1.51 MBytes
[ 7] 8.06-9.03      sec 23.8 MBytes  207 Mbits/sec  0  1.52 MBytes
[ 7] 9.03-10.00     sec 93.8 MBytes  810 Mbits/sec  0  1.57 MBytes
-----
[ ID] Interval      Transfer   Bitrate   Retr
[ 7] 0.00-10.00     sec 456 MBytes  382 Mbits/sec  118
[ 7] 0.00-10.01     sec 446 MBytes  374 Mbits/sec
sender
receiver

perf Done.
root@mininet-vm:/home/mininet#
s2-eth2 root

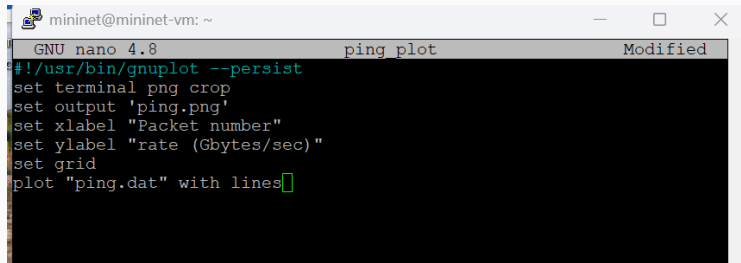
warning: this system does not seem to support IPv6 - trying IPv4
Server listening on 5201
Accepted connection from 10.0.0.1, port 42294
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 42296
[ ID] Interval      Transfer   Bitrate
[ 7] 0.00-1.00      sec 986 MBytes  8.27 Gbits/sec
[ 7] 1.00-2.01      sec 572 MBytes  4.75 Gbits/sec
[ 7] 2.01-3.03      sec 429 MBytes  3.55 Gbits/sec
[ 7] 3.03-4.00      sec 439 MBytes  3.77 Gbits/sec
[ 7] 4.00-5.00      sec 532 MBytes  4.46 Gbits/sec
[ 7] 5.00-6.00      sec 396 MBytes  3.32 Gbits/sec
[ 7] 6.00-7.00      sec 433 MBytes  3.63 Gbits/sec
[ 7] 7.00-8.00      sec 274 MBytes  2.29 Gbits/sec
[ 7] 8.00-9.01      sec 363 MBytes  3.01 Gbits/sec
[ 7] 9.01-10.00     sec 754 MBytes  6.42 Gbits/sec
[ 7] 10.00-10.00    sec 764 KBytes  5.25 Gbits/sec
-----
[ ID] Interval      Transfer   Bitrate
[ 7] 0.00-10.00     sec 5.06 GBytes  4.34 Gbits/sec
receiver

Server listening on 5201
Ciperf3: interrupt - the server has terminated
root@mininet-vm:/home/mininet# iperf3 -s
warning: this system does not seem to support IPv6 - trying IPv4
Server listening on 5201
Accepted connection from 10.0.0.1, port 57950
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 57956
[ ID] Interval      Transfer   Bitrate
[ 7] 0.00-1.00      sec 50.5 MBytes  423 Mbits/sec
[ 7] 1.00-2.04      sec 48.5 MBytes  390 Mbits/sec
[ 7] 2.04-3.00      sec 26.7 MBytes  233 Mbits/sec
[ 7] 3.00-4.00      sec 53.6 MBytes  451 Mbits/sec
[ 7] 4.00-5.00      sec 37.4 MBytes  314 Mbits/sec
[ 7] 5.00-6.00      sec 42.1 MBytes  353 Mbits/sec
[ 7] 6.00-7.07      sec 29.3 MBytes  229 Mbits/sec
[ 7] 7.07-8.07      sec 39.1 MBytes  330 Mbits/sec
[ 7] 8.07-9.03      sec 24.6 MBytes  214 Mbits/sec
[ 7] 9.03-10.00     sec 92.3 MBytes  798 Mbits/sec
[ 7] 10.00-10.01    sec 2.38 MBytes  1.46 Gbits/sec
-----
[ ID] Interval      Transfer   Bitrate
[ 7] 0.00-10.01     sec 446 MBytes  374 Mbits/sec
receiver
```

Figure 6: Объединение NETER и TBF

Выполнение лабораторной работы

```
mininet@mininet-vm: ~  
GNU nano 4.8 lab_netem_ii.py Modified  
Output: ping.dat  
***  
  
from mininet.net import Mininet  
from mininet.node import Controller  
from mininet.cli import CLI  
from mininet.log import setLogLevel, info  
import time  
  
def emptyNet():  
    "Create an empty network and add nodes to it."  
    net = Mininet(controller=Controller, waitConnected=True)  
  
    info('*** Adding controller\n')  
    net.addController('c0')  
  
    info('*** Adding hosts\n')  
    h1 = net.addHost('h1', ip='10.0.0.1')  
    h2 = net.addHost('h2', ip='10.0.0.2')  
  
    info('*** Adding switch\n')  
    s1 = net.addSwitch('s1')  
  
    info('*** Creating links\n')  
    net.addLink(h1, s1)  
    net.addLink(h2, s1)  
  
    info('*** Starting network\n')  
    net.start()  
  
    info('*** Set rate\n')  
    h1.cmdPrint('tc qdisc add dev h1-eth0 root tbf rate 10gbit burst 5000000 1')  
  
    time.sleep(10) # Wait 10 seconds  
  
    info('*** Starting iperf server on h2\n')  
    h2.cmdPrint('iperf3 -s &') # Launch server in foreground mode  
    info('*** Running iperf client from h1 to h2\n')  
    h1.cmdPrint('iperf3 -c ' + h2.IP() + ' | grep "Mbytes" | awk '{print $7}')  
  
    info('*** Stopping network')  
    net.stop()  
  
if __name__ == '__main__':  
    setLogLevel('info')  
    emptyNet()
```



The image shows a terminal window titled "mininet@mininet-vm: ~". Inside the terminal, the GNU nano 4.8 editor is open, editing a file named "ping_plot". The file content is a gnuplot script for generating a PNG plot. The script includes the following commands: `#!/usr/bin/gnuplot --persist`, `set terminal png crop`, `set output 'ping.png'`, `set xlabel "Packet number"`, `set ylabel "rate (Gbytes/sec)"`, `set grid`, and `plot "ping.dat" with lines`. The cursor is positioned at the end of the last line.

```
mininet@mininet-vm: ~  
GNU nano 4.8 ping_plot Modified  
#!/usr/bin/gnuplot --persist  
set terminal png crop  
set output 'ping.png'  
set xlabel "Packet number"  
set ylabel "rate (Gbytes/sec)"  
set grid  
plot "ping.dat" with lines
```

Figure 8: Скрипт для отрисовки графика

Выполнение лабораторной работы

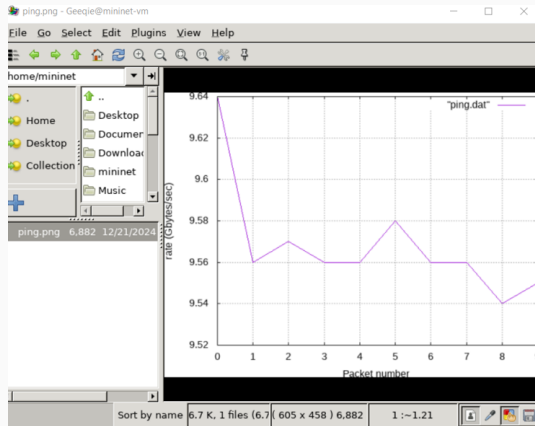


Figure 9: График изменения скорости передачи

В результате выполнения данной лабораторной работы я познакомился с принципами работы дисциплины очереди Token Bucket Filter, которая формирует входящий/исходящий трафик для ограничения пропускной способности, а также получил навыки моделирования и исследования поведения трафика посредством проведения интерактивного и воспроизводимого экспериментов в Mininet.

1. Mininet [Электронный ресурс]. Mininet Project Contributors. URL: <http://mininet.org/> (дата обращения: 17.11.2024).